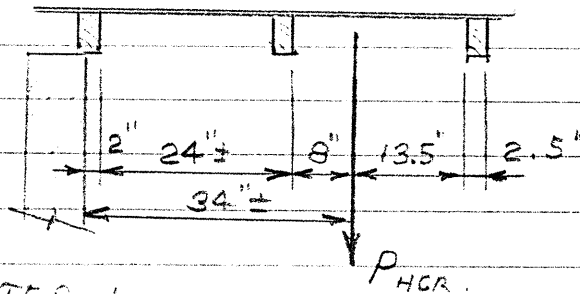


5 OF 6

## CHECK EXISTING RAFTERS FOR ADDED LOAD

MAX ADDITIONAL LOAD TO ONE RAFTER

$$\begin{aligned}
 P_{HCR} \text{ MAX} \\
 DL &= 263 \text{ \#} \\
 LL &= 455 \\
 TL &= 718
 \end{aligned}$$



MAX P TO ONE RAFTER:

$$\begin{aligned}
 P_{DL} &= (13.5/21.5) 263 = 165 \text{ \#} \\
 P_{LL} &= (13.5/21.5) 455 = 286 \text{ \#} \\
 P_{TL} &= (13.5/21.5) 718 = 451 \text{ \#}
 \end{aligned}$$

ASSUME MAX PL + LL ON HANGER + ROOF DL ONLY

RF - DL - TILE 8.0 \#/ft

JHTR 3.0

RAFTER 3.0

$$DL = 14.0 \text{ \#/ft} \rightarrow w_{DL} = 2 \times 14.0 = 28.0 \text{ \#/ft}$$

+ M<sub>MAX</sub> AT LOAD P<sub>MAX</sub>

$$\begin{aligned}
 + M_{MAX} &= 268 \times 8 - 28.0 \times 8^2/2 \\
 &= 1248 \text{ \#}
 \end{aligned}$$

$$\begin{aligned}
 + M_{MAX} &= 485 \times 2.8 - 28.0 \times 2.8^2/2 \\
 &= 1256 \text{ \#}
 \end{aligned}$$

FOR 3x6, S = 12.6

$$f_b = 12 \times 1256 / 12.6$$

$$= 1196 \text{ \#/ft} \text{ OK}$$

